

Machine Learning Presentation

Formula 1 Race Strategy Optimization

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Introduction to F1 and Machine Learning:

Formula 1 is the most data-intensive sport in the world. Every lap generates thousands of data points covering speed, throttle, brake pressure, tire temperature, fuel flow. Teams use this data to make split-second decisions that determine race outcomes.

300+

Sensors per Car

1.1M

Data Points / Second

24

Races per Season

Industry ML Adoption

Red Bull + Oracle

1B+ race simulations per weekend

Mercedes + AWS

Real-time ML strategy decisions

McLaren + Google Cloud

300M pre-race simulations

Ferrari + AWS HPC

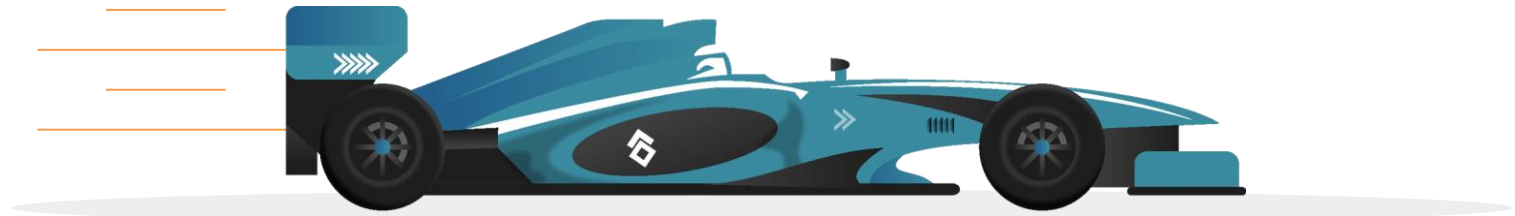
Car development acceleration



Problem Statement:

"Most existing research in Formula 1 analytics relies on historical race results and final standings to make predictions. However, this approach misses the detailed, lap-by-lap data that is generated during an actual race.

This project addresses that gap by working directly with real sensor and timing data recorded on every single lap, sourced from the official F1 live timing system via the FastF1 API.



Two Interconnected ML Problems:

ML Problem 01 — Lap Time Prediction

This is a **regression problem**, meaning the model's output is a continuous numerical value specifically, a driver's lap time in seconds.

The model takes in five inputs: **tyre compound, tyre age, fuel load, fuel correction, and speed trap readings.**

Using these variables, it learns to predict how long a lap will take under any given set of race conditions. The practical use case is that an engineer can feed in hypothetical scenarios and get a reliable lap time estimate, which then informs broader strategic decisions during a race weekend.

ML Problem 02 — Race Strategy Optimization

This combines two problem types: **classification and optimization.**

The classification component takes in real-time race variables like **tyre degradation rate, laps remaining, safety car status, and track position** and outputs a binary decision every single lap: **PIT or STAY OUT.**

The optimization component ensures that these per-lap decisions are not made in isolation, but are evaluated in terms of their cumulative effect on overall race time and final finishing position.

Dataset — FastF1 API (2023 Season):

Data Source: FastF1 Python Library

Official F1 live timing feed | Free, no API key | Returns pandas Data Frames directly | 7 seasons of data (2018–2024)

Circuit	Type	Key Characteristic	Why Selected
Bahrain	Technical	Baseline reference	Season opener: used in Paper 3 for direct comparison
Azerbaijan	Street Circuit	Highest safety car rate	Safety car every race 2018–2024, often multiple per race
Silverstone	Balanced	High tyre stress	Mixed corners, strong degradation signal
Monza	High Speed	Longest stints	Minimal degradation: contrasting strategy profile
Singapore	Street Circuit	Near-perfect SC rate	SC in every race except 2024, night race

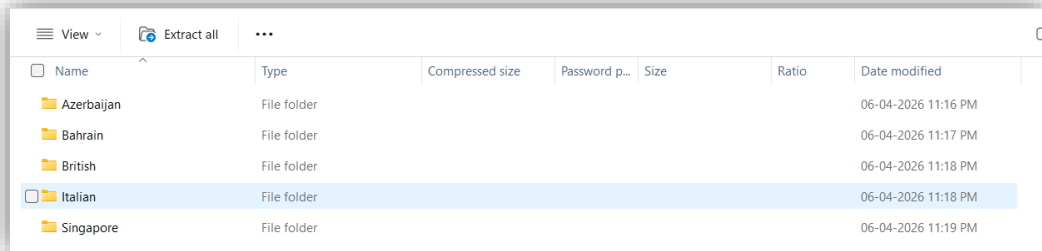
5,035 Total Rows

5 CSV Files per Circuit

2023 Season Only

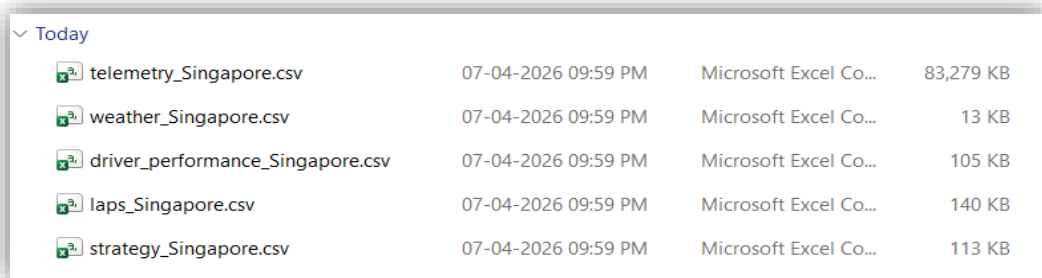
Real-World FastF1 Data

A View At the Data Sets:



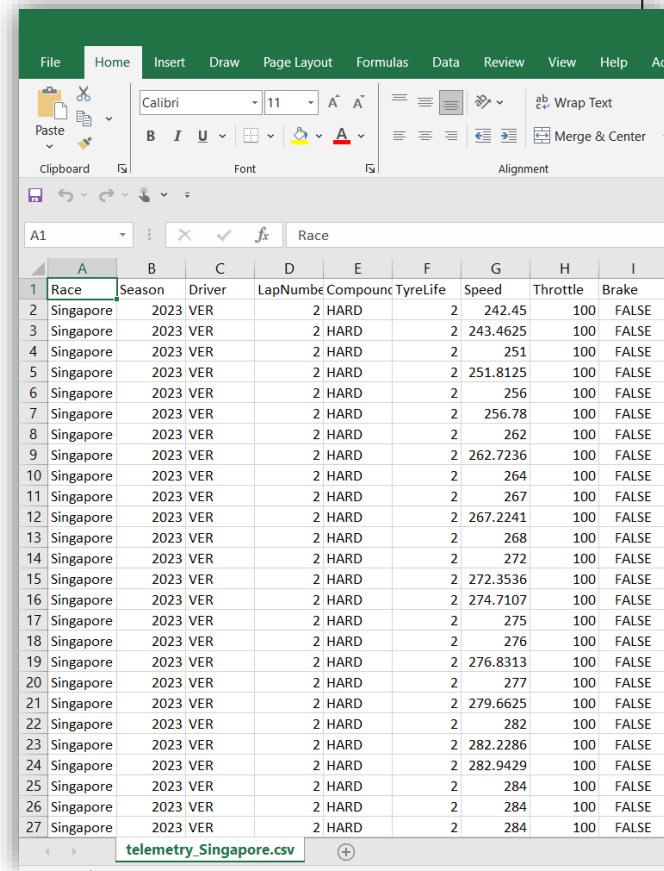
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Azerbaijan	File folder					06-04-2026 11:16 PM
Bahrain	File folder					06-04-2026 11:17 PM
British	File folder					06-04-2026 11:18 PM
Italian	File folder					06-04-2026 11:18 PM
Singapore	File folder					06-04-2026 11:19 PM

We have separated out the entire dataset into separate datasets for each individual race.



telemetry_Singapore.csv	07-04-2026 09:59 PM	Microsoft Excel Co...	83,279 KB
weather_Singapore.csv	07-04-2026 09:59 PM	Microsoft Excel Co...	13 KB
driver_performance_Singapore.csv	07-04-2026 09:59 PM	Microsoft Excel Co...	105 KB
laps_Singapore.csv	07-04-2026 09:59 PM	Microsoft Excel Co...	140 KB
strategy_Singapore.csv	07-04-2026 09:59 PM	Microsoft Excel Co...	113 KB

Every Race has all the information driver performance, total laps by each driver and strategy for each team.



Race	Season	Driver	LapNumber	Compound	TyreLife	Speed	Throttle	Brake
Singapore	2023	VER	2	HARD	2	242.45	100	FALSE
Singapore	2023	VER	2	HARD	2	243.4625	100	FALSE
Singapore	2023	VER	2	HARD	2	251	100	FALSE
Singapore	2023	VER	2	HARD	2	251.8125	100	FALSE
Singapore	2023	VER	2	HARD	2	256	100	FALSE
Singapore	2023	VER	2	HARD	2	256.78	100	FALSE
Singapore	2023	VER	2	HARD	2	262	100	FALSE
Singapore	2023	VER	2	HARD	2	262.7236	100	FALSE
Singapore	2023	VER	2	HARD	2	264	100	FALSE
Singapore	2023	VER	2	HARD	2	267	100	FALSE
Singapore	2023	VER	2	HARD	2	267.2241	100	FALSE
Singapore	2023	VER	2	HARD	2	268	100	FALSE
Singapore	2023	VER	2	HARD	2	272	100	FALSE
Singapore	2023	VER	2	HARD	2	272.3536	100	FALSE
Singapore	2023	VER	2	HARD	2	274.7107	100	FALSE
Singapore	2023	VER	2	HARD	2	275	100	FALSE
Singapore	2023	VER	2	HARD	2	276	100	FALSE
Singapore	2023	VER	2	HARD	2	276.8313	100	FALSE
Singapore	2023	VER	2	HARD	2	277	100	FALSE
Singapore	2023	VER	2	HARD	2	279.6625	100	FALSE
Singapore	2023	VER	2	HARD	2	282	100	FALSE
Singapore	2023	VER	2	HARD	2	282.2286	100	FALSE
Singapore	2023	VER	2	HARD	2	282.9429	100	FALSE
Singapore	2023	VER	2	HARD	2	284	100	FALSE
Singapore	2023	VER	2	HARD	2	284	100	FALSE
Singapore	2023	VER	2	HARD	2	284	100	FALSE

A view at how each dataset would look.

Literature Review:

Paper 1 — Championship Point Forecasting → Model 1 (Lap Time Prediction)

Key Finding: "This study tested seven machine learning algorithms on 74 years of Formula 1 data.

Gradient Boosting emerged as the best performing model, achieving an R^2 score of 0.999, meaning it could predict championship points with near-perfect accuracy. Feature importance analysis revealed that a driver's finishing position contributed 75.84% to the prediction, **confirming that race position is the single most influential factor in determining championship outcomes.**"

Paper 2 — From Data to Podium → Model 2 (Race Strategy Optimization)

Key Finding: "**This study built a machine learning model to predict whether a driver should pit or stay out on any given lap.** Among four algorithms tested, XGBoost performed best with an F1 Score of 0.7270.

The study also clustered all F1 drivers into four distinct groups based on their driving behavior, tactical decisions, and race performance proving that different driver styles require different pit stop strategies."

Model 1 Overview: Lap Time Prediction

REGRESSION PROBLEM

Predict a driver's lap time in seconds for any given race scenario.

Dataset Used: laps_[race].csv

Column	What it represents
LapNumber	Which lap in the race
Compound	Tyre type — SOFT, MEDIUM, HARD
TyreLife	How many laps on current tyres
FuelLoad_kg	Estimated fuel remaining
FuelCorrection_s	Time penalty due to fuel weight
Position	Driver's track position
SpeedI1	Speed trap 1 reading
SpeedI2	Speed trap 2 reading
SpeedFL	Finish line speed
SpeedST	Speed trap reading
Race	Circuit name — one-hot encoded
LapTime_s	Lap time in seconds — what we are predicting

Algorithms Used

Random Forest

Baseline Model

XGBoost Regressor

Primary Model

Data Filter Applied

IsAccurate = True removes pit laps, safety car laps, VSC laps from training data, o make sure the model only learns from laps where the driver was actually **racing at full speed** giving us a cleaner, more reliable prediction.

Evaluation Metrics

RMSE | MAE | R^2 (Coefficient of Determination)

Performance of Model 1:

An R^2 score of **0.9899** means the XGBoost model explains **98.99% of the variance** in lap times.

In practical terms, it has understood almost all of the underlying patterns that determine how fast a lap is.

Model 1 is production-viable. The predictions are accurate enough to directly feed into race strategy decisions with minimal correction.

Performance Comparison

Model	RMSE (seconds)	MAE (seconds)	R^2 Score
Random Forest (Baseline)	0.7299	0.3440	0.9890
XGBoost (Primary)	0.6978 ✓	0.3211 ✓	0.9899 ✓

98.99%

Lap Time Variance Explained

0.32s

Average Prediction Error

4,397

Clean Laps Trained On

The error is fairly negligible.

An average prediction error of **0.32 seconds** on an F1 lap that typically runs between 80–100 seconds is remarkably low.

if the model predicts a lap will take **91.50 seconds**, but the driver actually completes it in **91.82 seconds**, the prediction error for that lap is **0.32 seconds**.

The training size is honest, not inflated.

4,397 clean laps is a focused, quality-filtered dataset not an artificially large one.

Strong performance on a lean, clean dataset is actually more meaningful than the same numbers on a bloated one

Model 2 Overview: Race Strategy Optimization

BINARY CLASSIFICATION + OPTIMIZATION

Decide on every lap: should the driver PIT or STAY OUT?

Data Set Used: strategy_[race].csv

Column	What it represents
LapNumber	Which lap in the race
Compound	Tyre type — SOFT, MEDIUM, HARD
TyreLife	How many laps on current tyres
FuelLoad_kg	Estimated fuel remaining
FuelCorrection_s	Time penalty due to fuel weight
DegRate_s_lap	How much slower the tyre gets per lap
LapsRemaining	How many laps are left in the race
Position	Driver's current race position
SafetyCar	Whether safety car is active — 0 or 1
RedFlag	Whether red flag is active — 0 or 1
Race	Circuit name — one-hot encoded
IsPitLap	0 = Stay Out, 1 = Pit — what we are predicting

Class Imbalance Challenge

Stay Out (0): 4,812 laps — 97.02%

Pit (1): 148 laps — 2.98%

Imbalance Ratio: 32.5:1 — handled via scale_pos_weight

Algorithms

Decision Tree

Baseline

XGBoost + GridSearch

Primary + Tuned

Dynamic Programming — Breakeven Analysis

Formula: Pit is worth it when

$\text{DegRate} \times \text{LapsRemaining} > \text{Pit Loss Time}$

Performance of Model 2:

Accuracy alone is misleading here

Model	Accuracy	Precision	Recall	F1 Score
Decision Tree (Baseline)	0.7752	0.0894	0.7000	0.1585
XGBoost Tuned (Primary)	0.9294	0.2143	0.5000 ✓	0.3000 ✓

Why F1 Score matters more than Accuracy here:

A model predicting STAY OUT every single time would achieve 97% accuracy but would be completely useless. F1 Score balances Precision and Recall, penalizing models that miss pit laps.

The XGBoost model achieves an F1 Score of **0.30** compared to the Decision Tree's **0.1585**: nearly double. The Recall of **0.50** means the model correctly identifies half of all actual pit laps, which is a meaningful capability given how rare those events are in the data.

Dynamic Programming — Breakeven Analysis per Circuit

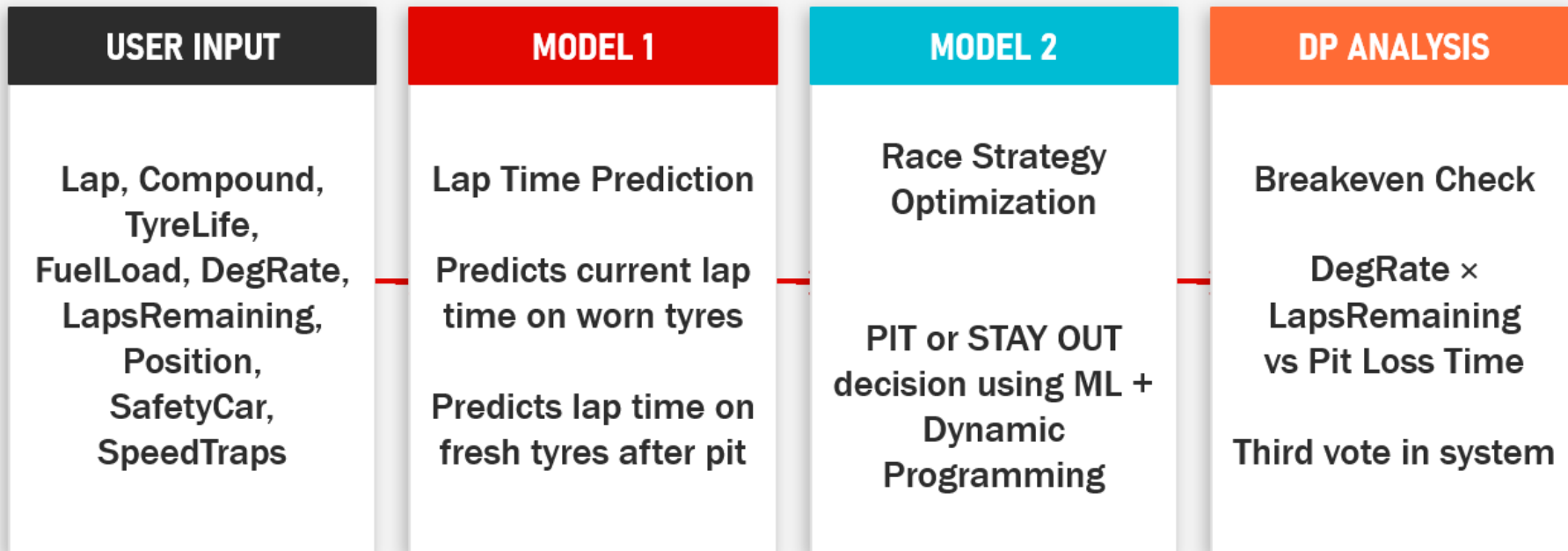
Circuit	Pit Loss Time	Avg Deg Rate	Breakeven (laps)	Time Gained	DP Decision
Azerbaijan	24.0s	0.2145 s/lap	111.9 laps	5.14s	STAY OUT
Bahrain	22.0s	0.2953 s/lap	74.5 laps	8.11s	STAY OUT
British	20.0s	0.2422 s/lap	82.6 laps	6.45s	STAY OUT
Italian	21.0s	0.2050 s/lap	102.4 laps	5.28s	STAY OUT
Singapore	28.0s	0.1954 s/lap	143.3 laps	5.95s	STAY OUT

Note: Average laps remaining across whole race (~27 laps) × avg deg rate (~0.23s) = ~6s gained < 20–28s pit loss → STAY OUT on average. Model correctly recommends PIT when deg rate and laps remaining are both high.

This is the **optimization layer** sitting on top of the classifier. It answers a more strategic question: *even if the model flags a lap as a potential pit window, does it actually make mathematical sense to pit?*

Unified Pipeline: How the Models Connect...

The two models are not isolated they form a sequential, interconnected pipeline



FINAL UNIFIED DECISION: Majority Vote (2 out of 3 signals) → PIT  or STAY OUT 

User Input Example:

We will now run the unified pipeline live using this race scenario:

DEMO INPUT SCENARIO		EXPECTED PIPELINE OUTPUT	
Race	Bahrain Grand Prix	Model 1 — Current Lap Time	~98–101s on worn SOFT
Lap Number	15 of 57	Model 1 — Post-Pit Lap Time	~95–97s on fresh tyre
Compound	SOFT	Lap Time Gain per Lap	~3–4s on fresh rubber
TyreLife	25 laps (heavily worn)	Model 2 — XGBoost Vote	PIT ✓
Fuel Load	80.0 kg	DP Check — Time Gained	$0.60 \times 42 = 25.2s > 22.0s$
Deg Rate	0.60 s/lap (high)	DP Decision	PIT ✓
Laps Remaining	42 laps	Safety Car Bonus	Free pit window — no time lost
Position	P3	Votes for PIT	3 / 3
Safety Car	YES ⚠		
Red Flag	No		
		FINAL DECISION: PIT NOW 	

Conclusion:

What We Set Out To Do

Applied machine learning to one of the most data-rich sports in the world which is Formula 1.

Addressed a real gap in existing research by working at the lap and telemetry level rather than race-result level.

What We Built

- Extracted and cleaned real-world data from the FastF1 API across 5 circuits of the 2023 season Bahrain, Azerbaijan, British, Italian and Singapore
- Built 5 structured CSV datasets per circuit covering laps, strategy, telemetry, driver performance and weather
- Trained and compared two algorithms for each model a baseline and a primary

Key Takeaway

What started as raw telemetry data from FastF1 became a working race strategy system that mirrors what billion-dollar F1 teams do with Oracle, Google Cloud and AWS built entirely using open-source Python tools

Thank You!

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